NOUREEN SIRAJ

WEB DEVELOPER

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PROFESSIONAL SUMMARY

Computer Science graduate (DHA Suffa University, Karachi) who builds complete web applications end to end — responsive front-ends in HTML, CSS and JavaScript, and Python back-ends with Flask and SQLite. Has independently taken ten projects through full development and testing cycles, spanning booking and reservation systems, e-commerce and productivity tools, and applied machine learning. Comfortable owning both the interface and the logic underneath it, with a strong eye for detail, clean code structure, and reliable, well-tested software.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL, HTML5, CSS3
Front-End	React (fundamentals), responsive design, DOM manipulation, state management
Back-End	Flask, Jinja2, Node.js and Express (fundamentals), RESTful APIs, server-side validation
Databases	SQLite, MySQL — schema design, CRUD operations, multi-table joins
Concepts	Object-Oriented Programming, Data Structures, Relational Database Design, Software Testing (manual)
Tools & Libraries	Git, GitHub, Tkinter, Netlify, MS Office (Excel, Word), JIRA (basic)
QA	Test case design, manual test execution, bug reporting, regression testing
Machine Learning	scikit-learn, TF-IDF feature extraction, Logistic Regression, Streamlit, joblib model persistence

FULL-STACK PROJECTS

Luna & Spice — Restaurant Table Reservations *Python · Flask · SQLite · Jinja2 · HTML · CSS*

- Built a table reservation system for a restaurant with capacity tracked per seating area per sitting, so bookings for the dining room, garden terrace, and chef's counter stop once that area's covers are full for the requested time.
- Modeled three seating areas with distinct capacities and party-size limits, and built an availability view showing covers remaining across every lunch and dinner sitting for the next available service days, automatically skipping the weekly closed day.
- Generated unique per-day booking references, validated guest name and phone number server-side, and re-checked seating capacity at submission time so two guests cannot claim the same covers after the page was first loaded.
- Built a reservation confirmation page and an admin booking list, plus a cancellation flow that releases covers back into that sitting's availability count immediately.

Gloss — Salon Chair Booking System *Python · Flask · SQLite · Jinja2 · JavaScript · HTML · CSS*

- Built a complete appointment booking system with duration-aware scheduling: each treatment consumes only the half-hour slots it actually needs, so a three-hour service blocks that stylist's chair for three hours and nothing double-books.
- Designed the SQLite schema and wrote the availability engine that calculates open start times per stylist, per day, per treatment length, and hides slots that would run past closing.
- Implemented server-side validation, flash messaging, a cancellation flow, and per-month booking reference generation.
- Rebuilt the same application as a dependency-free JavaScript version using browser storage, and deployed it live — demonstrating the same domain logic solved on both the server and the client.

WellCrest — Outpatient Appointment System *Python · Flask · SQLite · Jinja2 · HTML · CSS*

- Built an OPD booking system across five departments, with separate morning and evening clinic sessions on a twenty-minute slot grid and live counts of remaining slots per doctor.
- Enforced one-appointment-per-slot at the database level with a composite unique constraint, and handled the resulting integrity error in the booking route so two patients racing for the same slot never produce a double booking.
- Generated per-doctor, per-day appointment tokens (e.g. CARD-0831-01) and built the cancellation flow that releases a slot back into the schedule.

FRONT-END PROJECTS

DevHire — Developer Job Marketplace *HTML · CSS · JavaScript · React (basics)*

- Built the front-end of a hiring platform with role-based dashboards for developers and recruiters, plus a dynamic skill-based search UI.
- Optimized client-side filtering and rendering to keep candidate search fast and responsive as listings scaled.

ShopCart — E-Commerce Engine *HTML · CSS · JavaScript*

- Built the front-end of a digital retail engine with cart operations, live price calculation, and persistent session handling.
- Tested UI states across the checkout flow, catching edge cases in cart and pricing logic before release.

TaskFlow — Agile Kanban Dashboard *HTML · CSS · JavaScript · React (basics)*

- Built a state-synced Kanban board UI for real-time task and column management across a project workspace.
- Implemented smooth UI state updates for task actions using component state, keeping the interface responsive during data fetches.

Gloss — Salon Chair Booking System (Static Build) *HTML · CSS · JavaScript*

- Rebuilt the Gloss salon booking system as a dependency-free, front-end-only version, recreating the same duration-aware scheduling logic entirely in JavaScript with no backend.
- Used browser storage in place of a database to persist bookings and compute available slots per stylist, and deployed the build live.

Budget-Buddy — Personal Finance Tracker *HTML · CSS · JavaScript*

- Built the front-end for income and expense tracking with progress-bar budgets and live-updating spending charts.
- Implemented category-wise budget indicators and running-balance displays tied to live component state.

Quick Blog — Content Management System *HTML · CSS · JavaScript · React (basics)*

- Built the front-end for a CMS supporting article publishing, category filtering, and nested comment threads.
- Built a rich post editor UI with instant publishing and category-based content filtering.

MACHINE LEARNING — IN PROGRESS

Extending the development work above into applied machine learning — training models and serving live predictions through an interactive app.

Sentiment Analysis Web App *Python · scikit-learn · Streamlit · joblib*

- Trained a 3-class sentiment classifier (positive / neutral / negative) using TF-IDF feature extraction over unigrams and bigrams and a logistic regression model, evaluated with accuracy, a classification report, and a confusion matrix.
- Built a text-cleaning pipeline (lowercasing, URL/mention/number stripping, punctuation removal) applied consistently at both training and inference time.
- Persisted the trained model and vectorizer with joblib and built an interactive Streamlit interface where users type or pick an example sentence and get a live sentiment label with a confidence percentage.

PYTHON PROJECTS

AES File Encryption Tool *Python · cryptography · CLI*

- Built a CLI tool for AES symmetric encryption and decryption to protect sensitive files at rest.
- Added secure key derivation and exception handling for corrupted files and incorrect-key scenarios.

Password Manager *Python · AES Encryption · File I/O*

- Built a local, encrypted password vault unlocked with a single master password.
- Added input validation and fault-tolerant handling so a corrupted vault entry never crashes the app.

Scientific Calculator (GUI) *Python · Tkinter*

- Built a desktop scientific calculator supporting trigonometric, logarithmic, and power functions.
- Separated the UI layer from the math engine into reusable modules for easier testing and maintenance.

QR Code Generator *Python · qrcode library*

- Built a lightweight utility that generates QR codes from text, URLs, and contact details as image files.
- Structured the code for easy extension into batch processing and custom branding overlays.

SOFTWARE QUALITY ASSURANCE

Functional & Black-Box Testing

- Designed test matrices covering authentication, CRUD operations, form validation, and edge cases.
- Ran boundary-value and negative-assertion testing across every module in all six web development projects.
- Certified all six projects at sign-off with zero unresolved critical bugs after full test cycles.

API Testing & Automation

- Built Postman collections to validate HTTP responses, payloads, and status codes across REST endpoints.
- Wrote scripted assertions for JWT auth headers and token lifecycle checks within Postman environments.
- Logged structured JIRA bug reports with severity ratings and clear reproduction steps.

EDUCATION

Bachelor of Computer Science (BSCS)
DHA Suffa University, Karachi · 2022 – 2026

Intermediate, Pre-Engineering
Govt. Science and Commerce College, Block 07, Karachi · 2020 – 2022

CERTIFICATIONS

Python for Data Science, AI & Development
Coursera / IBM · 2025

Introduction to Front-End Development
Coursera · 2025

ADDITIONAL SKILLS & INTERESTS

Additional Skills	Front-End Development, Python Development, Software Quality Assurance (SQA), applied Machine Learning (in progress)
Soft Skills	Attention to Detail, Analytical Thinking, Problem Solving, Communication
Languages	English (Fluent), Urdu (Native)
Interests	Front-end development, Python development, software QA, database systems, machine learning, fintech, emerging technologies